BIG DATA PROJECT WORK | ABDA SKILL-GAP CAPSTONE
CHECKPOINT 2 – ONE-PAGE PROJECT PLAN

Team 9 | Amazon Data Engineer I (Job ID 10582732)

PROJECT: Retail Transaction Data Warehouse and Quality-Assured Analytics Pipeline

BUSINESS PROBLEM: Retail and fulfilment decision-makers need a consistent daily view of sales, product performance and customer purchasing patterns. Transaction data arriving in a raw format can contain duplicates, missing customer identifiers and invalid quantities or prices, making reporting unreliable. This project will create a repeatable analytical data pipeline and curated warehouse tables from historical retail transactions. The dataset is a retail-transaction proxy; it does not contain Amazon warehouse-level inventory, fulfilment-center or shipment-event data, so the project will not claim to measure warehouse operations.

DATASET & LICENSE: Online Retail II, UCI Machine Learning Repository. The dataset contains 1,067,371 transaction records from a UK-based online retailer (December 2009–December 2011), with invoice, product/stock code, description, quantity, invoice date, unit price, customer ID and country fields.
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Source: https://archive.ics.uci.edu/dataset/502/online+retail+ii

PLATFORM & APPROACH: Databricks Free Edition using Python, PySpark and Spark SQL. The source workbook will be downloaded on a laptop and uploaded to a Unity Catalog Volume; it will be converted to CSV before ingestion. Processing will be developed on a small sample first, then run on the full dataset subject to compute quota.

PIPELINE PLAN – EACH STEP IS LINKED TO A CHECKPOINT 1 GAP

Layer / step	Planned work and output	Checkpoint 1 gap addressed
Bronze – raw ingestion	Load source CSV files into a raw Delta table without business transformations; retain source columns and add ingestion timestamp and source-file metadata. Record input row count.	Big-data processing; warehouse/lakehouse foundations
Silver – transformation	Use PySpark to standardize data types and dates, remove exact duplicates, handle missing Customer ID transparently, and flag cancelled invoices and non-positive quantities/prices using documented rules. Preserve rejected/exception records for review.	Big-data/distributed processing; data modelling
Automated quality tests	Implement repeatable assertions for schema and required columns, duplicate keys, null rates, valid date/quantity/price ranges, and row-count reconciliation. Add transformation-level tests and an end-to-end run check; write pass/fail results to a quality log table.	Automated unit and integration testing
Gold – warehouse model	Create documented fact and dimension tables (e.g., FactSales, DimProduct, DimCustomer, DimDate and DimCountry) with defined keys and grain. Publish daily revenue/quantity by country and product, plus customer recency, frequency and spend summaries.	Data warehousing and warehouse-oriented modelling
Consumption	Use Spark SQL to query the gold tables and build a concise dashboard showing daily sales trend, leading products/countries and customer-value segments. Document the run order and rerun instructions.	Warehouse querying; maintainable, documented pipeline

SUCCESS CRITERIA & DELIVERABLES: The pipeline runs in sequence from raw to curated tables; key quality tests pass or have documented exceptions; warehouse tables have clear grain, keys and definitions; dashboard metrics reconcile to the cleaned data;

GOVERNANCE, LIMITATIONS & RISKS: Credit UCI and the CC BY 4.0 license. Treat Customer ID as a pseudonymous analytical key; do not attempt re-identification. Document missing identifiers, cancellations and any exclusions. This historical UK retailer dataset is not representative of Amazon operations and contains no warehouse or delivery events. Databricks Free Edition has daily compute limits and restricted notebook internet access; use a local download/upload workflow and test on a small sample before the full run.

LINK TO CHECKPOINT 1: The project directly targets Team 9's three stated priorities:
(1) automated unit/integration testing
(2) data warehousing and warehouse-oriented modelling
(3) big-data/distributed processing. PySpark/Spark SQL provide practical distributed transformation and querying; the plan does not claim experience with the preferred proprietary ETL products (Informatica, ODI, SSIS, BODI or DataStage).